

# Homework #6

CS5402 — Intro To Data Mining

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## 1 Support Vectors

The output is as follows:

```
Setting default kernel parameters
[[1]]
  X1 X2
2  2  1
3  4  2

[1] 1
[1] -1
```

## 2 Bayes Network

The network looks as follows:

The output looks as follows:

```
$location
location
Good  Bad
0.6  0.4
```

```
$quality
quality
  Good Normal  Bad
    0.3    0.5  0.2
```

```
$cost
, , location = Good
```

```
      quality
cost   Good Normal Bad
  High  0.8      0.6 0.1
  Low   0.2      0.4 0.9
```

```
, , location = Bad
```

```
      quality
cost   Good Normal  Bad
  High  0.6      0.6 0.05
  Low   0.4      0.4 0.95
```

```
$number_of_people
, , location = Good
```

```
      cost
number_of_people High Low
      High  0.6 0.8
      Low   0.4 0.2
```

```
, , location = Bad
```

```
      cost
number_of_people High Low
      High  0.1 0.6
      Low   0.9 0.4
```

```
prediction Good Normal
  Good      0      0
  Normal    1      1
  Bad       0      0
[1] Normal Normal
Levels: Good Normal Bad
```

### 3 DBScan

```
Clustering.DbscanResult(  
  [1, 5, 7, 12, 13, 14, 15, 16, 17, 22, 23, 24, 25, 29, 31],  
  [1, 1, 1, 1, 2, 1, 3, 1, 1, 1, 1, 4, 5, 6, 7, 8, 9, 1, 1, 1,  
    1, 10, 11, 12, 13, 1, 1, 1, 14, 1, 15, 1],  
  [18, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1])
```

Gadfly could not produce a graph.

### 4 Decision Tree

```
Feature 3, Threshold 118.0  
L-> Feature 3, Threshold 96.0  
  L-> Feature 3, Threshold 65.5  
    L-> Feature 1, Threshold 1.5  
      L-> 33.9 : 1/1  
      R-> Feature 3, Threshold 57.0  
        L-> 30.4 : 1/1  
        R-> 24.4 : 1/1  
    R-> Feature 3, Threshold 92.0  
      L-> Feature 1, Threshold 1.5  
        L-> 27.3 : 1/2  
        R-> 26.0 : 1/1  
      R-> 22.8 : 2/2  
  R-> Feature 1, Threshold 3.0  
    L-> Feature 3, Threshold 107.0  
      L-> Feature 2, Threshold 5.0  
        L-> 21.5 : 1/1  
        R-> 18.1 : 1/1  
      R-> Feature 4, Threshold 4.5  
        L-> 21.4 : 2/2  
        R-> 30.4 : 1/1  
    R-> 21.0 : 2/2  
R-> Feature 3, Threshold 192.5  
  L-> Feature 2, Threshold 7.0  
    L-> Feature 4, Threshold 4.5  
      L-> 17.8 : 1/2  
      R-> 19.7 : 1/1  
    R-> Feature 1, Threshold 2.5  
      L-> Feature 3, Threshold 162.5  
        L-> 15.2 : 1/2  
        R-> 19.2 : 1/2  
      R-> 16.4 : 1/3
```

```

R-> Feature 3, Threshold 237.5
  L-> Feature 3, Threshold 222.5
    L-> 10.4 : 2/2
    R-> 14.7 : 1/1
  R-> Feature 3, Threshold 254.5
    L-> 14.3 : 1/2
    R-> Feature 3, Threshold 299.5
      L-> 15.8 : 1/1
      R-> 15.0 : 1/1

```

## 5 KMeans

| Iters | objv         | objv-change   | affected |
|-------|--------------|---------------|----------|
| 0     | 1.000000e+02 |               |          |
| 1     | 4.880000e+01 | -5.120000e+01 | 0        |
| 2     | 4.880000e+01 | 0.000000e+00  | 0        |

K-means converged with 2 iterations (objv = 48.800000000000001)  
 Assignments: [1 1 1 1 2 1 3]  
 Counts: [5 1 1]